Build Your Research Portfolio Website From Scratch

A Complete Step-by-Step Guide Using Quarto + GitHub Pages

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1 Understanding the Basics

1.1 What Are We Building?

Before touching the keyboard, let's understand what we're doing in plain language. Your portfolio website is just a collection of text files on your computer. **Quarto** reads those files and converts them into a beautiful website. **GitHub** is a website where you store your files online — and it automatically shows them as a website to anyone in the world.

> **Key idea:** You write content in simple text files → Quarto turns them into a website → GitHub shows that website to the world — for free.

1.2 The Four Tools You Will Use

You will install and use exactly four tools:

Tool	What It Does
Git	Saves and tracks your files; sends them to GitHub
VS Code	A free text editor — you write all your content here
Quarto	Converts your text files into a beautiful website
GitHub	Free online service that hosts your website on the internet

1.3 What Is a Terminal / Command Prompt?

A terminal (also called command prompt) is a text-based way to control your computer. Instead of clicking buttons, you type commands. Don't be scared — you will only use about 6 simple commands in this whole guide.

How to open it on Windows: Press the Windows key → type `cmd` → press Enter. A black window opens. That's your terminal. Type commands there and press Enter to run them.

Important: Never close the terminal while a command is still running. Wait for the cursor (`>`) to appear again before typing the next command.

1.4 What Your Finished Website Will Look Like

Your portfolio will have these pages:

- **Home** — Your photo, name, bio, and research interests
- **About** — Detailed background, education, and experience
- **Research** — All your papers with links and personal notes
- **Blog** — Short research notes and field observations

Your website address will be: `https://yourusername.github.io` — free, forever.

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2 Installing the Tools (Windows)

You need to install three tools. Do them in order — do not skip ahead.

2.1 Install Git

Git is a tool that tracks every change you make to your files — like a time machine. It also lets you send your files to GitHub.

Step 1 — Open your browser and go to:

```
https://git-scm.com/download/win
```

Step 2 — The download will start automatically. If it doesn't, click the link that says "Click here to download manually".

Step 3 — Run the installer. Click **Yes** if Windows asks permission. Click **Next** → **Next** → **Next** → **Install** → **Finish**. Do NOT change any options. All the default settings are correct.

Step 4 — Verify it worked. Open a **NEW** terminal window and type:

```
git --version
```

You should see something like:

```
git version 2.44.0.windows.1
```

If you see "command not found": Restart your computer and try again.

2.2 Install VS Code

VS Code is a free text editor made by Microsoft. You will use it to write and edit all your website files.

Step 1 — Go to:

```
https://code.visualstudio.com
```

Step 2 — Click the big blue **Download for Windows** button. Run the downloaded installer.

Step 3 — Go through the installer. On the screen that says **"Select Additional Tasks"** — tick these two boxes:

- ✅ Add "Open with Code" action to Windows Explorer file context menu
- ✅ Add to PATH

Then click **Next** → **Install** → **Finish**.

Step 4 — VS Code will open automatically. Now install the Quarto extension:

- Look at the left side of VS Code — click the **Extensions** icon (4 small squares)
- In the search box, type: `Quarto`
- Click the first result (by quarto.org) → click **Install**

> **Tip:** The Quarto extension gives you syntax highlighting, live preview, and helpful autocomplete when writing your website files.

2.3 Install Quarto

Quarto is the tool that reads your text files and builds your actual website.

Step 1 — Go to:

```
https://quarto.org/docs/get-started
```

Step 2 — Click the download button for Windows. Run the file that downloads (ends in `.msi`).

Step 3 — Click through the installer with all default options.

Step 4 — **IMPORTANT:** Close your terminal and open a **NEW** one. Old terminal windows will not recognise newly installed tools.

Step 5 — Verify it worked:

```
quarto --version
```

You should see something like: `1.4.557`

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3 Setting Up GitHub

GitHub is the free service that will host your website. You need one account and one "repository" (a folder on GitHub where your files live).

3.1 Create Your GitHub Account

Step 1 — Go to:

```
https://github.com
```

Step 2 — Click **Sign up**. Fill in:

- Your email address
- A password
- A **username** — choose this carefully, it becomes part of your website address

> **Username tips:** Use your real name or a professional variation. No spaces allowed — use a dash or nothing. Example: `subhradip25` or `subhradip-bhattacharjee` Your website will be: `https://subhradip25.github.io`

Step 3 — GitHub will send a verification code to your email. Check your inbox and enter the code to activate your account.

3.2 Create Your Repository

A repository is like a project folder on GitHub.

Step 1 — After signing in, click the **+** icon at the top right → **New repository**

Step 2 — In the "Repository name" box, type **EXACTLY**:

```
yourusername.github.io
```

Replace `yourusername` with your actual GitHub username. For example: `subhradip25.github.io`

Step 3 — Make sure **Public** is selected (not Private).

Step 4 — Scroll down and click the green **Create repository** button.

> **Why this special name?** The name `yourusername.github.io` tells GitHub to automatically serve this repository as a live website at `https://yourusername.github.io` — completely free!

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4 Creating Your Quarto Project

Now we create the actual website project on your computer.

4.1 Create the Project

Step 1 — Open VS Code. Click **Terminal** in the top menu → **New Terminal**. A panel opens at the bottom. This is your terminal inside VS Code.

Step 2 — Navigate to your Documents folder:

```
cd Documents
```

Step 3 — Create a new Quarto website project:

```
quarto create project website myportfolio
```

Quarto will ask one question about the directory name. Just press **Enter** to accept.

Step 4 — Move into the project folder:

```
cd myportfolio
```

Step 5 — Open the project in VS Code:

```
code .
```

> **What does `code .` mean?** The dot (`.`) means "open the current folder". VS Code will open with all your project files listed in the left panel.

4.2 Understanding Your File Structure

After creating the project, you will see these files on the left side of VS Code:

File	Purpose
`_quarto.yml`	Master settings — menu, title, theme, output folder
`index.qmd`	Your home page — bio, photo, research interests
`about.qmd`	Your detailed about/CV page
`styles.css`	Custom colours and design tweaks

4.3 Preview Your Site Locally

Before making any changes, let's see what your site looks like right now.

Step 1 — In the VS Code terminal, type:

```
quarto preview
```

Step 2 — Wait 10–15 seconds. Your browser will automatically open and show your website. This is your **local preview** — only visible on your computer.

Step 3 — When you want to stop the preview, press **Ctrl+C** in the terminal.

> **Live reload:** While the preview is running, every time you save a file (Ctrl+S), the browser automatically refreshes to show your changes. You don't need to restart the preview each time.

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5 Building Your Home Page

The home page is the first thing visitors see — your name, photo, bio, and research interests.

5.1 Edit _quarto.yml — Site Settings

This file controls your site's title, menu structure, and visual theme. Click on `_quarto.yml` in the left panel, **select all text** (Ctrl+A), **delete it**, and paste this:

```yaml
project:
  type: website
```

```yaml
  output-dir: docs

website:
  title: "Your Full Name"
  navbar:
    left:
      - href: index.qmd
        text: Home
      - href: about.qmd
        text: About
      - href: research.qmd
        text: Research
      - href: blog.qmd
        text: Blog

format:
  html:
    theme: cosmo
    css: styles.css
    toc: true
```

Press **Ctrl+S** to save. Your browser preview will update automatically.

> **Tip:** You can change the `theme` from `cosmo` to other options like `flatly`, `litera`, `sandstone`, or `journal` to change the look of your site.

5.2 Edit index.qmd — Home Page

Click on `index.qmd`, **select all**, **delete**, and paste this template. Then fill in your own details:

```
---
title: "Your Full Name"
subtitle: "Geospatial & Agronomy Researcher"
image: images/profile.png
about:
  template: jolla
  links:
    - icon: envelope
      text: Email
      href: mailto:your@email.com
    - icon: mortarboard-fill
      text: Google Scholar
      href: https://scholar.google.com/your-profile
    - icon: file-person
      text: ResearchGate
      href: https://researchgate.net/your-profile
```

```
    - icon: linkedin
      text: LinkedIn
      href: https://linkedin.com/in/yourprofile
---

I am a researcher at **Your Institution**, specialising in
geospatial analysis and agronomy.

My work focuses on understanding agricultural systems using
remote sensing, GIS, and field-based approaches.

## Research Interests

- Geospatial analysis of agricultural landscapes
- Coastal agronomy and soil health
- Remote sensing for crop monitoring
- Climate impacts on farming systems
```

Press **Ctrl+S** to save and check your browser.

5.3 Add Your Profile Photo

Step 1 — Create an `images` folder in the VS Code terminal:

```
mkdir images
```

Step 2 — Open Windows File Explorer. Navigate to:

```
Documents → myportfolio → images
```

Copy your photo into this folder and **rename it** to exactly: `profile.png` (or `profile.jpg` if your photo is a JPG file)

Step 3 — In `index.qmd`, make sure this line matches your file extension:

```
image: images/profile.png   # for PNG files
image: images/profile.jpg   # for JPG files
```

Step 4 — Save and check your browser. Your photo will appear in a circle.

> **Photo tips:** Use a clear, well-lit, professional photo. Portrait or square orientation works best.

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6 Building Your Research Page

The research page is the most important page for a researcher. It lists all your publications with links, co-authors, and your personal plain-language notes.

6.1 Create research.qmd

In VS Code, click **File** → **New File** → name it `research.qmd` → save it inside your `myportfolio` folder.

6.2 Format for Each Paper

Use this template for each paper. Copy and repeat it for every publication:

```
---
title: "Research"
---

## Journal Articles

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**Your Paper Title Here** (2024)
*Journal Name* | Co-authors: Name A, Name B
[Read Paper](https://doi.org/your-doi-link) | [ResearchGate](https://researchgate.net)

> **In simple words:** Write 2–3 lines here explaining what this paper
> is about and why it matters. This is your personal note for visitors
> — not the abstract.

---
```

6.3 Export All Papers from Google Scholar

If you have many papers, don't type them manually. Use this shortcut:

Step 1 — Go to your Google Scholar profile and sign in.

Step 2 — Tick the **checkbox** at the top to select all papers. A blue bar will appear.

Step 3 — Click **Export** → select **BibTeX** → a `.bib` file downloads with all your paper details.

Step 4 — Open claude.ai, paste the BibTeX content, and ask:

> *"Format all these papers into a research.qmd file for a Quarto portfolio"*

Claude will generate the complete, properly formatted page instantly.

> **Suggested structure:** Organise by type: Journal Articles → Book Chapters → Conference Papers → PhD Thesis. Put newest papers first within each section.

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7 Building Your Blog

The blog is where you share short research notes, field observations, and methodology explainers — content that is valuable but not suited for a formal paper.

7.1 Create the Blog Structure

Step 1 — Create `blog.qmd` in your `myportfolio` folder with this content:

```
---
title: "Blog"
listing:
  contents: posts
  sort: "date desc"
  type: default
  categories: true
---
```

Step 2 — Create the `posts` folder in the terminal:

```
mkdir posts
```

Step 3 — Create a folder for your first post:

```
mkdir posts\first-post
```

Step 4 — In VS Code, right-click the `first-post` folder ⟶ **New File** ⟶ name it `index.qmd`

7.2 Write Your First Blog Post

Paste this template into `posts/first-post/index.qmd`:

```
---
title: "Your Post Title Here"
date: "2025"-05-01
categories: [remote sensing, agronomy]
---

Write your post content here in plain English.

You can use:

- Bullet points
- **Bold text**
- *Italic text*

Add an image like this:
![Caption text](../../images/your-image.png)
```

Blog ideas for researchers: Field observations • Methodology notes • Literature review summaries • Conference takeaways • Software tutorials • Data analysis tips • Explaining your paper in simple language

7.3 Adding More Posts Later

For every new blog post, follow the same pattern:

```
# Create a folder for the new post
mkdir posts\your-post-name

# Then in VS Code, create: posts/your-post-name/index.qmd
# Fill in title, date, categories, and content
```

> **Auto-sorted:** Quarto automatically sorts posts by date (newest first) and creates a category filter. You don't need to do anything extra.

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8 Publishing Your Site Live

This is the exciting part — making your site publicly visible on the internet. This is a one-time setup. After this, updating takes just 4 commands.

8.1 Step 1 — Tell Git Who You Are

Run these two commands in the VS Code terminal. Use your own email and name:

```
git config --global user.email "your@email.com"
git config --global user.name "Your Full Name"
```

8.2 Step 2 — Build the Final Website

This creates the `docs/` folder containing all your website files:

```
quarto render
```

> **What happens:** Quarto reads all your `.qmd` files and builds the complete website into a folder called `docs/`. You will see this folder appear in your VS Code file panel on the left.

8.3 Step 3 — Connect Your Folder to GitHub

Run these two commands (replace `yourusername` with your actual GitHub username):

```
git init
git remote add origin https://github.com/yourusername/yourusername.github.io.git
```

8.4 Step 4 — Upload to GitHub

```
git add .
git commit -m "My portfolio first version"
git push -u origin main
```

Authentication: A browser window may open asking you to authorise Git Credential Manager. Click **"Authorise git-ecosystem"**. This is completely safe and only happens once.

8.5 Step 5 — Configure GitHub Pages

Step 1 — On GitHub.com, open your repository.

Step 2 — Click **Settings** in the top menu.

Step 3 — In the left sidebar, click **Pages**.

Step 4 — Under "Build and deployment":

- Source → **Deploy from a branch**
- Branch → **main**
- Folder → **/docs**
- Click **Save**

Step 5 — Wait 2–3 minutes, then visit:

```
https://yourusername.github.io
```

You're live! Your site is now publicly accessible — free, forever, no subscription, no ads.

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9 Updating Your Site

After your site is live, updating it is very simple. Every time you add or change content, run these four commands:

9.1 The 4-Command Update Workflow

```
quarto render
git add .
git commit -m "Describe what you changed"
git push
```

Your site will update on the internet within 1–2 minutes. Refresh your browser to see the changes.

9.2 Common Update Tasks

Task	What to do
Add a new blog post	Create `posts/name/index.qmd` → write → run 4 commands
Update your bio	Edit `index.qmd` → save → run 4 commands
Add a new paper	Edit `research.qmd` → add entry → save → run 4 commands
Change your photo	Replace `images/profile.png` → run 4 commands
Fix a typo	Edit the file → save → run 4 commands

Memorise these 4 commands: `quarto render` → `git add .` → `git commit -m "message"` → `git push`

These 4 commands are all you need for the rest of your career to keep your portfolio updated.

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10 Adding Images and Figures

Quarto supports simple images, scientific figures with captions, interactive maps, and live charts from Python or R code.

10.1 Simple Photo or Image

Save your image file in the `images/` folder, then add one line to your `.qmd` file:

```
![Caption for your image](images/my-photo.jpg)
```

To control the display size:

```
![My field survey in Goa](images/field.jpg){width=80%}
```

To align to the right:

```
![Caption](images/photo.jpg){width=40% fig-align="right"}
```

10.2 Side-by-Side Image Comparison

Perfect for before/after maps, satellite image comparisons, or treatment effects:

```
::: {layout-ncol=2}
![Before treatment](images/before.jpg)

![After treatment](images/after.jpg)
:::
```

You can also do a 3-column layout:

```
::: {layout-ncol=3}
![2021](images/map2021.jpg)

![2022](images/map2022.jpg)

![2023](images/map2023.jpg)
:::
```

10.3 Scientific Figure with Caption and Cross-Reference

For publication-quality figure referencing — exactly like in a research paper:

```
::: {#fig-ndvi}
![](images/ndvi-map.png)

NDVI map of the study area, Kharif season 2023.
Green = healthy vegetation, red = stressed/bare areas.
:::

As clearly shown in @fig-ndvi, the spatial distribution of
crop stress follows the drainage pattern of the region.
```

Auto-numbering: The `@fig-ndvi` reference automatically creates a numbered, clickable figure link. Quarto numbers all your figures automatically — just like LaTeX in a research paper.

10.4 Interactive Map (Python / Folium)

For geospatial researchers, you can embed a fully interactive map that visitors can zoom and click:

```python
#| label: fig-map
#| fig-cap: "Study sites, ICAR-CCARI research area, Goa"

import folium
m = folium.Map(location=[15.49, 73.82], zoom_start=11)

folium.Marker(
    [15.49, 73.82],
    popup="ICAR-CCARI, Goa",
    tooltip="Click for details"
).add_to(m)

m
```

Requirement: Python and folium must be installed. Install folium with: `pip install folium`

10.5 Charts from Python

Paste your existing Python code directly into your `.qmd` file. Quarto runs it and shows the chart automatically — no export needed:

```python
#| label: fig-yield
#| fig-cap: "Soybean yield trend, coastal Goa research plots"

import matplotlib.pyplot as plt

years = [2019, 2020, 2021, 2022, 2023]
yield_data = [2.1, 2.4, 2.0, 2.8, 3.1]
```

```
plt.figure(figsize=(8, 4))
plt.plot(years, yield_data, marker='o', color='#1D9E75', linewidth=2)
plt.fill_between(years, yield_data, alpha=0.1, color='#1D9E75')
plt.title("Soybean yield trend, Goa")
plt.xlabel("Year")
plt.ylabel("Yield (tonnes/ha)")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

10.6 Embedding a QGIS Map

Export your QGIS map as an interactive HTML file using the `qgis2web` plugin, then embed it in your portfolio:

Step 1 — In QGIS: Plugins ⟶ qgis2web ⟶ Export to Leaflet ⟶ Save as HTML

Step 2 — Copy the exported folder into your project (e.g., `maps/my-map/`)

Step 3 — Embed it in any `.qmd` file:

````
```{=html}
<iframe src="maps/my-map/index.html"
        width="100%"
        height="500px"
        frameborder="0">
</iframe>
````

```

---

## 11 Quick Reference Card

### 11.1 Commands You Will Use Most

| Command | What it does |
|---|---|
| `quarto preview` | Start local preview in browser |
| `quarto render` | Build the complete website into docs/ |
| `git add .` | Stage all changed files |
| `git commit -m "message"` | Save a snapshot with a description |
| `git push` | Upload to GitHub (site goes live) |
| `mkdir foldername` | Create a new folder |
| `cd foldername` | Move into a folder |
| `code .` | Open current folder in VS Code |

## 11.2 Markdown Cheat Sheet

| What you type | What appears |
|---|---|
| `**bold text**` | **bold text** |
| `*italic text*` | *italic text* |
| `## Heading` | A section heading |
| `- item` | Bullet point |
| `[link text](https://url.com)` | Clickable link |
| `![alt](images/photo.jpg)` | An image |
| `` `code` `` | Inline code |

## 11.3 Useful Links

| Resource | Link |
|---|---|
| Quarto documentation | https://quarto.org/docs |
| Your live portfolio | https://yourusername.github.io |
| Your GitHub repository | https://github.com/yourusername/ yourusername.github.io |
| Google Scholar export | https://scholar.google.com |
| ORCID profile | https://orcid.org |

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*Portfolio: https://subhradip25.github.io*

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